Assignment 1

◄ Tutorial for Assignment 1 on N-Grams　　　Assignment 2 ►

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Assignment 1
by Srija Mukhopadhyay - Sunday, 14 January 2024, 1:04 AM

Hey all! ヾ(ˆ ∇ ˆ)

Assignment 1 on Language Modelling through Smoothing and Interpolation is out now!

Please go through the requirements document to find out the complete details about the same. In case you have any concerns, please feel free to reach out to the TAs.

In addition to that, there will be a tutorial soon where we will take more doubts and walk you through a few essential components that might help you with the assignment and the course.

Please follow the submission instructions carefully. The deadline for the assignment is 28th January, 23:59. This is a hard deadline. Please start early and plan accordingly. Hope you have fun doing the assignment!

Cheers,

iNLP TAs

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Re: Assignment 1
by Harshita Harshita - Wednesday, 17 January 2024, 11:12 AM

Can you give an example for the output for tokenizer and also specify the input format of the corpus_path, like as a command line argument or as an input variable.

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Re: Assignment 1
by Hardik Mittal - Wednesday, 17 January 2024, 11:00 PM

1) **Input**: "In 'Pride and Prejudice' by Jane Austen, Elizabeth Bennett meets Mr Darcy at a ball hosted by her friend @charles_bingly. They dance, but Mr Darcy finds her behaviour "tolerable, but not handsome enough to tempt him" #rude. She later visits Pemberley, Mr Darcy's estate, where she learns more about his character. Check out more information at https://janeausten.co.uk."

Output: List of Lists [[List of tokenized words of sentence 1] , [List of tokenized words of sentence 2] ...]

Example : [["In", "'", "Pride", "and", "Prejudice", "'", "by", "Jane", "Austen", ',', 'Elizabeth', 'Bennett', 'meets', 'Mr', 'Darcy', 'at', 'a', 'ball', 'hosted', 'by', 'her', 'friend', '<MENTION>', '.'] , ['They', 'dance', ',', 'but', 'Mr', 'Darcy', 'finds', 'her', 'behavior', '"', 'tolerable', ',', 'but', 'not', 'handsome', 'enough', 'to', 'tempt', 'him', '"', '<HASHTAG>', '.'] , ['She', 'later', 'visits', 'Pemberley', ',', 'Mr', 'Darcy', "'s", 'estate', ',', 'where', 'she', 'learns', 'more', 'about', 'his', 'character', '.'] , ['Check', 'out', 'more', 'information', 'at', '<URL>', '.']]

Your tokenizer is up to you; you can use it in a variety of ways. As you browse the dataset, we encourage you to handle any additional cases you come across, such as stage marking, sound effects, scene descriptions, etc.

2) Command line argument to the path of the corpus you want to use.
Example : ```**python3 language_model.py <lm_type> <corpus_path>**```
[PS : Just copied it from the assignment doc itself.]

?

Hope your queries are cleared, if not, feel free to reply back to this thread!

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Re: Assignment 1
by Animesh Das - Friday, 19 January 2024, 3:44 AM

For collecting the corpus_path from cmd line, can we rename the files... 'Pride and Prejudice - Jane Austen.txt', 'Ulysses James Joyce.txt' ?

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Re: Assignment 1
by Ayan Datta - Friday, 19 January 2024, 9:39 PM

You may rename the files. Make sure the filenames or paths are not hardcoded anywhere in your code.

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Re: Assignment 1
by Radhika Garg - Wednesday, 17 January 2024, 12:01 PM

one more question in addition to this as Input should be a txt corpus file path in cmd or it should be a sentence as it is specified in submission format?
And with respect to tagging, How exactly it should be done or how it should be highlighted? Can u give an example with input and Expected O/P

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Re: Assignment 1
by Hardik Mittal - Wednesday, 17 January 2024, 11:05 PM

1) Yes, the input should be a command line argument to text corpus file path (just import the downloaded datasets) and then use the sentence tokenizer to break the text file into sentences

2) **Input**: "In 'Pride and Prejudice' by Jane Austen, Elizabeth Bennett meets Mr Darcy at a ball hosted by her friend @charles_bingly. They dance, but Mr Darcy finds her behaviour "tolerable, but not handsome enough to tempt him" #rude. She later visits Pemberley, Mr Darcy's estate, where she learns more about his character. Check out more information at https://janeausten.co.uk."

Output: List of Lists [[List of tokenized words of sentence 1][List of tokenized words of sentence 2] ...]

Example : [["In", "'Pride", "and", "Prejudice'", "by", "Jane", "Austen", ',', 'Elizabeth', 'Bennett', 'meets', 'Mr', 'Darcy', 'at', 'a', 'ball', 'hosted', 'by', 'her', 'friend', '<MENTION>', '.'] , ['They', 'dance', ',', 'but', 'Mr', 'Darcy', 'finds', 'her', 'behavior', '"', 'tolerable', ',', 'but', 'not', 'handsome', 'enough', 'to', 'tempt', 'him', '"', '<HASHTAG>', '.'] , ['She', 'later', 'visits', 'Pemberley', ',', 'Mr', 'Darcy', "'s", 'estate', ',', 'where', 'she', 'learns', 'more', 'about', 'his', 'character', '.'] , ['Check', 'out', 'more', 'information', 'at', '<URL>', '.']]

Hope your queries are cleared, if not, feel free to reply back to this thread!

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Re: Assignment 1
by Sanya Gandhi - Thursday, 25 January 2024, 11:04 AM

My code cosiders (') as different token
So, my output is:
[['In', "'", 'Pride', 'and', 'Prejudice', "'", 'by', 'Jane', 'Austen', ',', 'Elizabeth', 'Bennett', 'meets', 'Mr', 'Darcy', 'at', 'a', 'ball', 'hosted', 'by', 'her', 'friend', '', '.'], ['They', 'dance', ',', 'but', 'Mr', 'Darcy', 'finds', 'her', 'behaviour', '"', 'tolerable', ',', 'but', 'not', 'handsom
'enough', 'to', 'tempt', 'him', '"', '', '.'], ['She', 'later', 'visits', 'Pemberley', ',', 'Mr', 'Darcy', "'", 's', 'estate', ',', 'where', 'she', 'learn.
'more', 'about', 'his', 'character', '.'], ['Check', 'out', 'more', 'information', 'at', '', '.']]

I don't see this affecting further parts of the assignment. Is this considered valid?

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Re: Assignment 1
by Sanya Gandhi - Thursday, 25 January 2024, 11:08 AM

```
tokenized text: [['In', '"', 'Pride', 'and', 'Prejudice', '"', 'by', 'Jane', 'Austen', ',', 'Elizabeth', 'Bennett', 'meets', 'Mr', 'Darcy', 'at', 'a', 'ball', 'hosted', 'by', 'her', 'friend', '<MENTION>', '.'], ['They', 'dance', ',', 'but', 'Mr', 'Darcy', 'finds', 'her', 'behaviour', '"', 'tolerable', ',', 'but', 'not', 'handsome', 'enough', 'to', 'tempt', 'him', '"', '<HASHTAG>', '.'], ['She', 'later', 'visits', 'Pemberley', ',', 'Mr', 'Darcy', "'", 's', 'estate', ',', 'where', 'she', 'learns', 'more', 'about', 'his', 'character', '.'], ['Check', 'out', 'more', 'information', 'at', '<URL>', '.']]
```

IDK why tokens are being removed from the post. But it's there.

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Re: Assignment 1
by Ayan Datta - Saturday, 3 February 2024, 10:52 PM

Considering to split ' as a different token is alright.

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Re: Assignment 1
by Sagnick Bhar - Thursday, 18 January 2024, 11:03 AM

1). In N=3-gram linear interpolation case, how do I find the probabilty of a sentence?
For example,
sentence: “I am eating mango”
P(“I am eating mango”) = [λ1*P("eating" | "I am") + λ2*P("eating" | "am") + λ3*P("eating")] * [λ1*P("mango" | "am eating") + λ2*P("mango" | "eating") + λ3*P("mango")]
Is this correct or is it something else?

2). Should we use start tag(~~) and end tag(~~) for each sentence or is it okay to omit?
For example: In bigram,
[["~~", "'I", "am", "eating'", "mango", "~~"]]

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Re: Assignment 1
by Hardik Mittal - Thursday, 18 January 2024, 2:51 PM

1) You need to add the EOS and BOS tags. After adding them, the equation will look like the following :

$$
\begin{aligned}
P(\text{"I am eating mango"}) = & [\lambda_1 \cdot P_{tri}(\text{"I"}|\text{"<START> <START>"}) + \lambda_2 \cdot P_{bi}(\text{"I"}|\text{"<START>"}) + \lambda_3 \cdot P_{uni}(\text{"I"})] \\
& \cdot [\lambda_1 \cdot P_{tri}(\text{"am"}|\text{"<START> I"}) + \lambda_2 \cdot P_{bi}(\text{"am"}|\text{"I"}) + \lambda_3 \cdot P_{uni}(\text{"am"})] \\
& \cdot [\lambda_1 \cdot P_{tri}(\text{"eating"}|\text{"I am"}) + \lambda_2 \cdot P_{bi}(\text{"eating"}|\text{"am"}) + \lambda_3 \cdot P_{uni}(\text{"eating"})] \\
& \cdot [\lambda_1 \cdot P_{tri}(\text{"mango"}|\text{"am eating"}) + \lambda_2 \cdot P_{bi}(\text{"mango"}|\text{"eating"}) + \lambda_3 \cdot P_{uni}(\text{"mango"})] \\
& \cdot [\lambda_1 \cdot P_{tri}(\text{"<END>"}|\text{"eating mango"}) + \lambda_2 \cdot P_{bi}(\text{"<END>"}|\text{"mango"}) + \lambda_3 \cdot P_{uni}(\text{"<END>"})]
\end{aligned}
$$

2) Yes, you should add (N-1) <BOS> tags (where 'N' represents the number of words in your N-gram) and a single <EOS> tag to the sentence. This ensures that the probability of each word is accounted for, providing a more consistent probability distribution across all N-grams in the model. By experimenting with both approaches, you should ideally observe an improvement in model performance when these tags are included.

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Re: Assignment 1
by Soham Ghosh - Thursday, 25 January 2024, 9:52 PM

In linear interpolation, while estimating lambda values by the method given in the paper it is possible that one or two lambda values may remain 0 at the end. So we keep those lambda values as it is while calculating probabilities?

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Re: Assignment 1
by Ayan Datta - Saturday, 3 February 2024, 7:37 PM

Yes you are required to keep the lambda values and use them as calculated using the algorithm.

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Re: Assignment 1
by Vaishnavi Khule - Thursday, 18 January 2024, 6:15 PM

3
N-Grams
Design a function in Python that takes a value of 'N' and the and generates an N-sized N-gram model from both the given corpus.
for doing this ,should we consider the tokenised output from previous task or should we write one other independent function?

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Re: Assignment 1
by Ayan Datta - Friday, 19 January 2024, 2:24 PM

You are required to use the tokenizer built in the assignment for the ngram model. Taking the corpus path or the tokenized sentences as input to the function is a design decision left to you.

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Re: Assignment 1
by Nivesh Mittapally - Saturday, 20 January 2024, 10:04 PM

In Good Turing approach, what should be done for the missing values of frequency and the most frequent element.

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Re: Assignment 1
by Ayan Datta - Saturday, 3 February 2024, 7:37 PM

You are required to use linear regression to interpolate between the known values of frequencies. You may refer to the Tutorial Slides for more details.

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Re: Assignment 1
by Aryan Verma - Monday, 22 January 2024, 4:35 PM

1. Using the generated N-gram models (without the smoothing techniques),
try generating sequences. Experiment with different values of N and
report which models perform better in terms of fluency.
2. Attempt to generate a sentence using an Out-of-Data (OOD) scenario
with your N-gram models. Analyze and discuss the behavior of N-gram
models in OOD contexts.
In the generation part, we were asked to try these 2 things. without using the smoothing technique. My doubt here is "as we are storing ngrams in the dictionary and we are appending the corresponding next words as list of values wherever this ngram appeared earlier.". So if this particular value had appeared earlier with the same key in the corpus, it would predict. but say a case, where an input sentence is totally different from the tokens stored, it would definitely not predict any next word as we are not using any smoothing technique". is this the same which we need to try?

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Re: Assignment 1
by Vithesh Reddy Adala - Friday, 26 January 2024, 4:52 AM

3 N-Grams

Design a function in Python that takes a value of 'N' and the <corpus_path> and generates an N-sized N-gram model from both the given corpus.

What do you mean by N-gram **model?** Is it just the list of all n-grams? The function is supposed to return a list of all n-grams from the given corpus right?

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Re: Assignment 1
by Ayan Datta - Thursday, 1 February 2024, 7:22 PM

The N-gram model may be of whatever data type you choose to represent it in (An instance of a class, a list, or a dictionary).

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Re: Assignment 1
by Srija Mukhopadhyay - Friday, 26 January 2024, 4:58 PM

Hey all! /ᐠ .ᆺ. ᐟ\ﾉ

The deadline for assignment 1 has been extended to 2nd February, 23:59. Please note that no further extensions will be possible after that.

Cheers,
INLP TAs

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Re: Assignment 1
by Animesh Das - Wednesday, 31 January 2024, 7:10 PM

In the assignment pdf it is mentioned:
1. Using the generated N-gram models (without the smoothing techniques), try generating sequences.
...
3. Now try to generate text using the models with the smoothing techniques (LM1, LM2, LM3, LM4)

Source Code for Generation:
`generator.py `
LM type can be g for Good-Turing Smoothing Model and i for Interpolation Model.

For the without smoothing case, how do you expect to run the evaluations?

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Re: Assignment 1
by Ayan Datta - Thursday, 1 February 2024, 7:11 PM

You may use any letter for the ngram with no smoothing. Make sure to mention such instructions for running your code in your README.

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Re: Assignment 1
by Nivesh Mittapally - Thursday, 1 February 2024, 9:40 PM

Can you give an example for probability of a sentences using trigram good turing smoothing.

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Re: Assignment 1
by Ayan Datta - Saturday, 3 February 2024, 8:09 PM

Sentence: I like apples

$$P(\text{I like apples}) = P(\text{I}|\text{<START><START>}) * P(\text{like}|\text{<START>I}) * P(\text{apples}|\text{I like}) * P(\text{<END>}|\text{like apples})$$

You may refer to the tutorial slides regarding calculating the individual probabilities.

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Re: Assignment 1
by Ayan Datta - Friday, 2 February 2024, 10:54 AM

Hey all,
The formula for the probability of a word, given its history using Good-Turing Smoothing, has been updated in the tutorial slides. The denominator has been changed to use the Good-Turing estimate to calculate the count of the history and shall now handle unseen contexts. We apologize for any inconvenience caused.
With regards to the change made in the tutorial slides and extension requests, the deadline for the assignment has been extended to 5 February, 11:59 pm. Please note that the deadline shall not be extended any further.

Cheers,
iNLP TAs

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Re: Assignment 1
by Nivesh Mittapally - Friday, 2 February 2024, 1:17 PM

So if the denominator is sum(c*(w1, w2, w')) where w' is every possible vocabulary. Then if (w1, w2, w') pair has frequency zero then we have to take c*=N1 right? (just want to clarification)

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Re: Assignment 1
by Ayan Datta - Saturday, 3 February 2024, 8:11 PM

Yes, you may use c*=N1 for the assignment. You are free to experiment with ways to estimate the value of N0.

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Re: Assignment 1
by Pradhyumna Palore - Friday, 2 February 2024, 11:01 AM

In Good Turing, the probability of unseen is defined as Nr1/Nr
Where Nr1 is frequency of elements coming once
Q1. Is this Nr original Nr or not
Q2. If it is original Nr, the frequency of elements coming 1 will generally be very high, may be around 0.6/0.7 also
So is this good thing to provide such high probability for unseen

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Re: Assignment 1
by Pradhyumna Palore - Saturday, 3 February 2024, 1:39 PM

kindly clarify the above doubt

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Re: Assignment 1
by Ayan Datta - Saturday, 3 February 2024, 10:46 PM

As given in the wikipedia resource:
The probability of an unseen species can be taken to be p0 = N1/N. Where N is the number of observed distinct species.

So the Probability of an NGram will be P(w1w2w3) = Count*(w1w2w3)/N where Count* is the adjusted count and N is the number of observed distinct ngrams
We know that **P(w3|w1w2) = P(w1w2w3)/P(w1w2) = P(w1w2w3)/sum(P(w1w2wi) for all wi in vocabulary) = Count*(w1w2w3)/sum(Count*(w1w2wi) for all wi in vocabulary) as N cancels out**
This implies that the **Probability of an unseen word given a context will be p_0/P(w1w2), or as N cancels out Nr1/sum(Count*(w1w2wi) for all wi in vocabulary) and not N1/N**
Note that the context may be unknown, in which case the Count*(w1w2wi) becomes Nr

A1) You may use either the S(Nr) or the original Nr for the unseen case.
A2) As per the resources given unseen words do get a high probability. You may experiment with different estimations for p0 but for the assignment you may stick to the estimation provided in the wikipedia page. Please note that any other estimation methods will require justification and proper derivation of the formula used.

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Re: Assignment 1
by Sriya Sriya - Sunday, 4 February 2024, 11:36 PM

Hi TAs,
Please confirm if its ok that we submit some intermediate .py files (linearinterpolation.py , goodturingsmoothing.py) as well apart from 3 files which are mentioned in the assignment pdf (language_model.py, tokenizer.py, generator.py) , the length of the languagemodel.py file would become quite large if we put the logic of linear interpolation, good turing smoothing and without smoothing techniques all in one file, it might become hard to debug and be less readable.

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Re: Assignment 1
by Saheli Saha - Monday, 5 February 2024, 9:25 AM

TAs please confirm this as we have to start finalising our files for submission.

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Re: Assignment 1
by Ayan Datta - Monday, 5 February 2024, 7:14 PM

Yes, it is alright. In fact, you are encouraged to use intermediate .py files, as they help make the code more modular and help with readability.

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Re: Assignment 1
by Vithesh Reddy Adala - Friday, 2 February 2024, 1:15 PM

Can we submit ipynb files instead of py? Because it's much easier to show different individual experiments with the use of cells. Thus, I request you to allow the submission of .ipynb files as well.

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Re: Assignment 1
by Jainit Bafna - Monday, 5 February 2024, 7:34 PM

Certainly, you can submit the .ipynb files, **but only as intermediary files**. But **Please ensure that the .py files we requested are included** because we'll be running your code in the **terminal for evaluation and plagarism checks.** Usually, we don't allow .ipynb files because they tend to be large due to the outputs they contain.

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Re: Assignment 1
by Nivesh Mittapally - Saturday, 3 February 2024, 2:41 PM

Can you tell me what is the difference between README and the Report that needs to be submitted? And also can you tell what exactly has to be there in both of them.

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Re: Assignment 1
by Ayan Datta - Monday, 5 February 2024, 7:11 PM

Reports generally contain all your observations (for ex, graphs, and metrics) regarding and answers to the questions posed in the assignment, whereas the README contains instructions on how to run your code (requirements and commands), any model or data URLs, and any assumptions used in your implementation.

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Re: Assignment 1
by Soham Ghosh - Monday, 5 February 2024, 3:01 PM

Can we keep the 2 corpus files and the 4 language model files in the submission folder?

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Re: Assignment 1
by Ayan Datta - Monday, 5 February 2024, 7:07 PM

Submitting the corpus files isn't necessary, regardless, you may do so as long as you stay within the file size limits. If you exceed the limit, you may attach a drive link in your README.

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◄ Tutorial for Assignment 1 on N-Grams

Assignment 2 ►